Muqeeth Mohammed

personal-web muqeeth101@gmail.com

Education

- **University of Montréal** Montréal, QC
  PhD in Computer Science — Advisor: Prof. Aaron Courville *Sep 2024 - current*
  - **Research Interests**: Multiagent Reinforcement Learning, Game Theory, LLMs, PEFT, MoEs
- **University of North Carolina - Chapel Hill** Chapel Hill, NC
  Masters of Science in Computer Science— Advisor: Prof. Colin Raffel *Aug 2021 - July 2023*
- **Indian Institute of Technology Madras** Chennai, India
  Bachelor of Technology (Honors) in Electrical Engineering — GPA:9.36/10.0 (Rank 8/60) *Jul 2016 – May 2020*

Experience

- **MILA Quebec AI Institute** Montréal, QC
  PhD student— Advisor: Prof. Aaron Courville *Sep 2024 - current*
  - Designed a social dilemma testbed for LLMs and demonstrated that standard RL robustly converges to greedy behavior
  - Scaled opponent learning awareness algorithms to LLMs to learn reciprocity in two player general sum games.
- **MIT-IBM** Cambridge, MA
  Research Engineer — Manager: Rameswar Panda *Jan 2024 - Aug 2024*
  - Worked on adapting large scale mixture of experts models (MoEs) to improve performance on poorly representated domains in their pretraining data.
- **University of North Carolina - Chapel Hill** Chapel Hill, NC
  Research Assistant — Advisor: Prof. Colin Raffel *Aug 2021 - Dec 2023*
  - Designed a post-hoc routing scheme among specialized experts to enable zero-shot generalization to unseen tasks.
  - Designed a new method SMEAR that enables standard gradient based training for mixture of experts models, outperforming all gradient estimators and heuristic routing with no increase in compute costs compared to discrete routing.
  - Co-led the development of *T-Few*, a few-shot fine-tuning method widely adopted and highly cited (1k+)
- **Indian Institute of Technology, Delhi** Delhi, India
  Research Assistant — Advisors: Prof. Mausam, Prof. Soumen Chakrabarti *Jul 2020 - Jul 2021*
  - Identified inconsistencies when translating OpenIE data from English to other languages and built a system that overcame these inconsistencies to provide state-of-the-art performance on the OpenIE task over five languages.
  - Created and publicly released OpenIE test sets for low-resource Indian languages like Hindi and Telugu.
- **Senior Thesis** Chennai, India
  Undergrad Student — Advisors: Prof. Sutanu Chakraborti, Prof. Anil Prabhakar *Jan 2020 - May 2020*
  - Designed a retrieval system to retrieve relevant courses to a given query course from the university courses at IITM.
  - Used explicit semantic analysis with the addition of phrases to achieve 0.868 nDCG score compared to a neural model baseline of 0.735. Implemented the idea of retrofitting to user feedback to further improve the score to 0.92.

MISCELLANEOUS

- Received ESP Excellence scholarship at University of Montreal
- Received Masters Merit Fellowship covering tuition and stipend at UNC Chapel Hill
- Recevied RamShriram Merit Scholarship at IIT Madras for maintaining 8.0+ GPA every year
- Reviewer: ICML2026, ICLR2025, ICML2024, NeurIPS 2023, ACL 2022, EMNLP 2022
- Selected for and attended Cooperative AI Foundation Summer School, 2025 and 2026